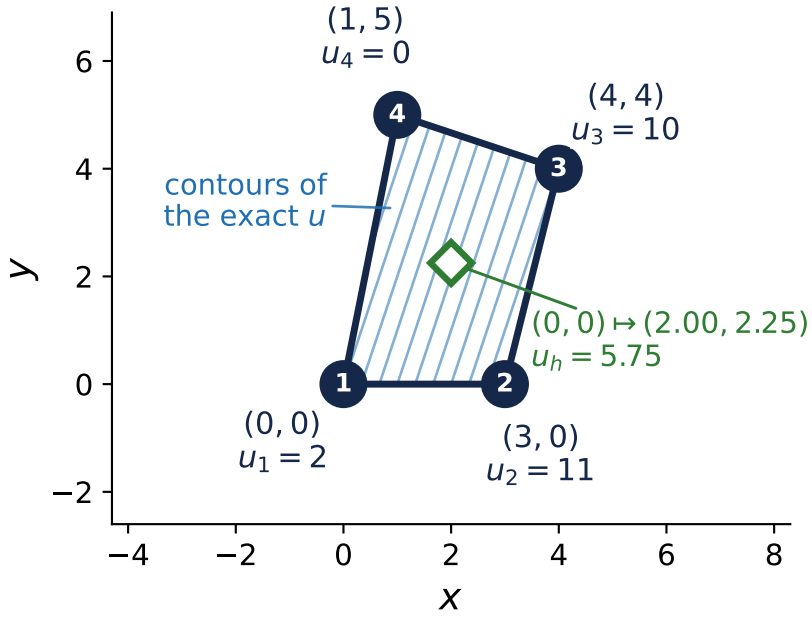
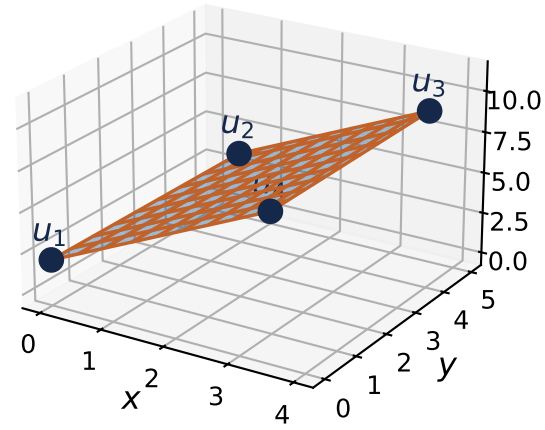
Same element, same four nodes, same rule $u_h = \sum_a N_a u_a$ --- only the sampled field changes.

A. the four numbers the element is given
$u_a = u(x_a, y_a)$ --- read off the blue contours

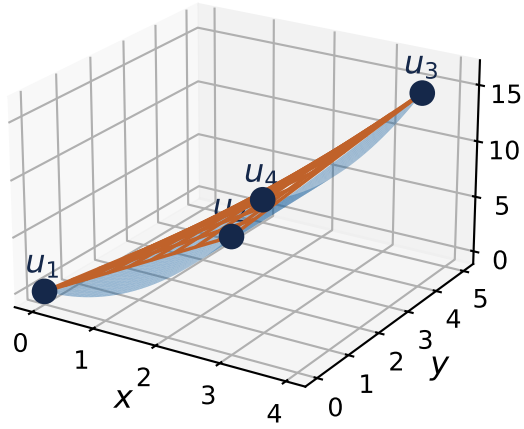


B. a LINEAR field, $u = 2 + 3x - y$



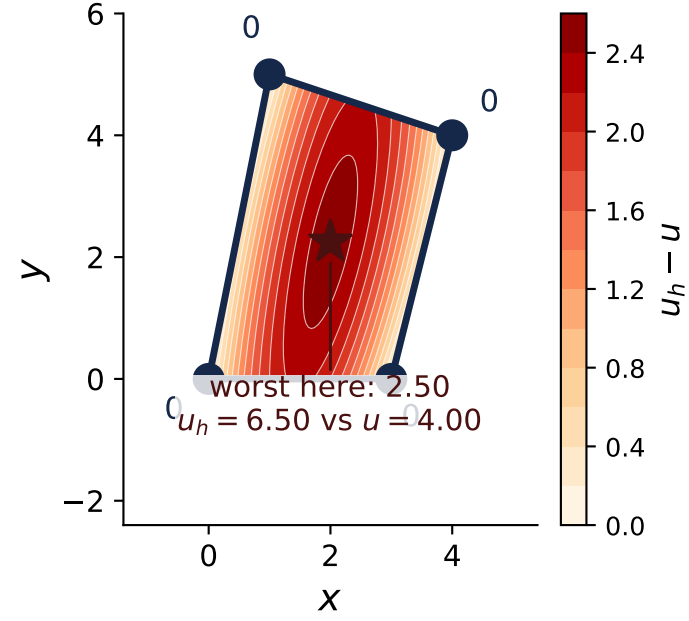
the two surfaces coincide: $u_h = u$
$\max|u - u_h| = 3.6e-15$

C. a CURVED field, $u = x^2$



the flat sheet cannot bend: $u_h \neq u$
$\max|u - u_h| = 2.500$

D. panel C's miss, mapped
zero at the nodes, worst in between



u: a function, defined everywhere --- what you are after. u_a: four numbers, one per node.

$u_h = \sum_a N_a u_a$: the function the element rebuilds out of them. In B it equals u; in C and D it does not.